ARISTON

HEAT PUMP WATER HEATER

EVERY DAY ONE SHOWER IS FREE

The future is already here

It heats the water using the air it offers comfort respecting the environment.

For many years Ariston has chosen a green path and for this reason it has been investing in the most advanced technology of its sector. Thanks to the research, we have developed the new heat pump water heaters NUOS & AR-Series. They are efficient and capable of heating a high quantity of water with an extremely low consumption of electricity, as they absorb the heat directly from the external air.

NUOS heat pump water heater range is available both in monoblock and split version (with an external unit), and can be installed both on the wall and on the floor, with a capacity from 80 up to 300 litres and for the AR-Series with varies types 6PM, 10PM, 17PTP and the highest 35PTP.
In the new buildings NUOS range & AR-Series can be used as an alternative to the solar thermal as renewable energy source.

	MONOBLOCK	SPLIT
WALL HUNG	NUOS 80-100-120 / NUOS PRIMO 80-100	
FLOOR STANDING	NUOS 200-250-250 SYS / NUOS PRIMO 200-240-240SYS / NUOS PLUS 200-250-250SYS 250TWIN SYS	NUOS EVO SPLIT 300

nuos

SINGLE PHASE	THREE PHASE
AR - 6PM	AR - 17PTP
AR - 10PM	AR - 35PTP

AR-Series

NUOS range
Monoblock or Split?

Ariston offers a complete range of split and monoblock heat pump water heaters	SPLIT	MONOBLOCK
Integration with internal rooms	• The internal unit is more compact because it doesn't have the refrigeration unit • The gas connections have extremely limited dimensions • The condensate discharge is not necessary	
Architectonic integration		The monoblock heat pump outside needs only discharge grates that don't have any visual impact.
Air treatment		The monoblock product allows to use the cold and dehumidified air for cooling the internal rooms through appropriate canalizations.
Sound emission	Fan and compressor are the main responsible for the sound emission in an aerothermal heat pump. Thanks to the external unit, the product doesn't cause any noise inside the installation room	
Trasport	It is usually easier to transport two smaller and lighter parcels (internal and external unit).	
Installation	The external unit installation needs minimum wall works. It is enough ø50 mm hole for the refrigerant gas pipes.	
Initial start-up		The initial start-up is immediate as it is a plug&play product. The operator doesn't need refrigerator technician's skills and equipment.
Maintenance	The ordinary maintenance main operation of an aerothermal heat pump consists in cleaning the evaporator. This operation is easier in the external unit (if it is in a place easy to reach).	

AR-Series model
Single phase atau three phase?

THREE PHASE		SINGLE PHASE		Ariston offers a model of heat pump for hot water needs in a big quantity
/ AR-35PTP	/ AR-17PTP	/ AR-10PM	/ AR-6PM	MODEL
				Comfort
Apartment / Hotel / Hospital Industrials / School		Villa / Bungalow	Residential house	Hot water used
• Compressor high temperature protection. • Compressor High & Low pressure protection. • Compressor over-load protection. • High water temperature protection. • Motor overcurrent & overheat protection. • Power supply shortage protection. • Anti-freezing protection. • Ambient temp sensor .		• Defrosting temperature sensor. • Water flow switch failure protection. • Power-off memory function & Automatic restart function. • Intelligent defrosting function.. • Smart water compensation.		Safety
4 COP				Coeffisient of Performance COP
		• Heat pump single phase has a low noise level around 56-58 dB (A)		Noise
		• AR-Series type 6 kW and 17 kW have a circulating pump installed in the unit, so it does not require an external pump.		Circulation pump

NUOS / AR-Series, the product range that fits any need

	NUOS PRIMO 80-100	NUOS 200-250-250SYS	NUOS PRIMO 200-250-250SYS	NUOS PLUS 200-250-250SYS 250 TWINS	NUOS EVO SPLIT 300	AR-6PM AR-10PM	AR-17PTP AR-35PTP
Guide to the choice by application							
New Single and Two–family house	●	●	●	●	●	●	●
New Multi-family house	●	●	●	●		●	●
Exiting Single and Two-family house	●	●	●	●	●		
Existing Multi-family house	●						
Apartment	●					●	●
Hospital	●	●	●	●	●		
School	●				●		
Shop	●	●	●	●	●	●	
Industrial / factory		●	●	●		●	●
Hotel		●	●	●		●	●

● typical and advisable application ● possible application

NUOS range for all the applications

NUOS PRIMO

Wall-hung heat pump water heater

PERFORMANCES

- **COP 3,0** with air temperature at 20° C (EN 255-3)
- Working range in heat pump mode with air temperature from 10° C to 40° C
- Ecologic gas **R134A** allows to reach water temperature up to 55° C in heat pump mode ON
- Condenser wrapped around the tank
- Titanium enamelled steel tank
- Integrated heating element
- Anti-Corrosion anode in magnesium
- LED Module display
- Green, Boost, Auto, Anti-Legionella function and time setting of the widrawal

Wall-hung installation available in 80, 100 liters

TECHNICAL DATA		80	100
COP air 20°C water 15-55°C (EN 255-3)		3,0	3,0
Min/max air temperature	°C	10/40	10/40
Thermal power with air at 20°C (*)	W	750	750
Average absorbed power (*)	W	250	250
Heating time with air at 20°C (*)	h,min	4,45	6,00
Max. quantity of hot water at 40°C (*)	l	110	141
Storage tank capacity	l	80	100
Thermal losses in 24h	W	0,46	0,5
Max. operating pressure	bar	8	8
Voltage/ max. power consumption (*)	V/W	220/1450	220/1450
Heating element power	W	1200	1200
Standard air flow rate	m^3/h	170	170
Min. volume of the installation room (**)	m^3	20	20
Sound pressure level at 1 m	dB(A)	38	38
Weight when empty	Kg	45	49
Protection rating		IP24	IP24
Insulation thickness	mm	31	31
Diameter of hydraulic connections	"	1/2M	1/2M
Min. temperature of the storage tank room	°C	1	1

(*) Water temperature 15-55°C with air at 20°C (EN 255-3)

(**) Room minimum volume in case of installation without canalization

	NUOS PRIMO 80	NUOS PRIMO 100
CODE	3629006	3629007

NUOS PRIMO FLOOR STANDING

COMFORT
- Green, boost, auto, antilegionella functions and time setting of the withdrawals
- Coil and probe-holder for solar and/or boiler integration
- Additional heating element

EFFICIENCY&ENERGY SAVING
- COP 3,1 with air temperature at 20°C (EN 255-3)
- COP 2,8 with air temperature at 7°C (EN 255-3)
- Ecologic gas R134A allows to reach water temperatures up to 62°C in heat pump mode
- Working range in heat pump mode with air temperatures from -5 to 42°C.

QUALITY
- Active anode (protech) + magnesium anode
- Titanium enamelled steel tank
- Condenser wrapped around the tank

DESIGN
- Led module display

CE

Easiness of use, -5°C inlet air min. temperature. Renewable product

TECHNICAL DATA		NUOS 200	NUOS 240	NUOS 240 SYS
COP 7°C (EN 16147)		2,41	2,64	2,64
COP 7°C (EN 255-3)		2,8	2,8	2,8
COP 20°C (EN 255-3)		3,1	3,1	3,1
Heating time with air at 7°C (EN 16147)	h:min	7:45	9:33	09:33
Min/max air temperature	°C	-5/42	-5/42	-5/42
Sound power	db(A)	53	53	53
Average absorbed power (*)	W	500	500	500
Max. quantity of hot water at 40°C (**)	l	284	348	348
Storage tank capacity	l	200	242	242
Max. operating pressure	bar	6	6	6
Voltage/ max. power consumption (*)	V/W	220-230/2750	220/2670	220/2670
Heating element power	W	2000	2000	2000
Standard air flow rate	m^3/h	400	400	400
Minimum volume of installation room (***)	m^3	20	20	20
Weight when empty	kg	87	92	107
Protection rating		IP24	IP24	IP24
Insulation thickness	mm	35	35	35
Diameter of hydraulic connections	"	3/4 M	3/4 M	3/4 M
Min. temperature of storage tank room	°C	1	1	1
Coil surface	m^2	-	-	0,65

(*) Water temperature 15-55°C with air at 20°C (EN 16147)
(**) Water temperature 15-62°C with air at 20°C (EN 16147)
(***) Minimum room volume in case of installation without canalization

CODE			
	3079008	3079009	3079010

NUOS PLUS

NEW 2015

COMFORT
- Auto, boost, green, voyage, boost2, antilegionella functions and time setting of the withdrawals
- 1 or 2 coils and probe-holder for solar, boiler or biomass integration
- Additional double power steatite heating element

EFFICIENCY&ENERGY SAVING
- COP 3,7 with air temperature at 20°C (EN 255-3)
- COP 3,1 with air temperature at 7°C (EN 255-3)
- Ecologic gas R134A allows to reach water temperatures up to 62°C in heat pump mode
- Working range in heat pump mode with air temperatures from -7 to 42°C
- Photovoltaic function integrated

QUALITY
- Active anode (protech) + magnesium anode
- Titanium enamelled steel tank
- Condenser wrapped around the tank

DESIGN
- LCD display
- Integrated multi-diameter adapter for air inlet and outlets
- Integrated handling straps

Maximum efficiency and easy integration. A+ ready for 2017 ErP standard.

TECHNICAL DATA		NUOS PLUS 200	NUOS PLUS 250	NUOS PLUS 250 SYS	NUOS PLUS 250 TWIN SYS
COP 7°C (EN 16147)		Available from November 2015			
Heating time with air at 7°C (EN 16147)	h:min	Available from November 2015			
Min/max air temperature	°C	-7/42	-7/42	-7/42	-7/42
Sound power	dB(A)	Available from November 2015			
Average absorbed power (*)	W	700	700	700	700
Max. quantity of hot water at 40°C (**)	l	Available from November 2015			
Storage tank capacity	l	200	250	250	250
Max. operating pressure	bar	6	6	6	6
Voltage/ max. power consumption (*)	V/W	220-240/2500	220-240/2500	220-240/2500	220-240/2500
Heating element power	W	1500+1000	1500+1000	1500+1000	1500+1000
Standard air flow rate	m^3/h	650	650	650	650
Minimum volume of installation room (***)	m^3	30	30	30	30
Weight when empty	kg	90	95	115	130
Protection rating		IPX4	IPX4	IPX4	IPX4
Insulation thickness	mm	50	50	50	50
Diameter of hydraulic connections	"	3/4 M	3/4 M	3/4 M	3/4 M
Min. temperature of storage tank room	°C	1	1	1	1
Coil surface	m^2	-	-	0,65	0,65+0,65

(*) Water temperature 15-55°C with air at 20°C (EN 16147)
(**) Water temperature 15-62°C with air at 20°C (EN 16147)
(***) Minimum room volume in case of installation without canalization

CODE				
	3079052	3079053	3079054	3079055

AVAILABLE FROM NOVEMBER 2015

NUOS

COMFORT

- Auto, boost, green, voyage, boost2, antilegionella functions and time setting of the withdrawals
- Coil and probe-holder for solar and/or boiler integration
- Additional double power steatite heating element

EFFICIENCY&ENERGY SAVING

- COP 3,7 with air temperature at 20°C (EN 255-3)
- COP 3,1 with air temperature at 7°C (EN 255-3)
- Ecologic gas R134A allows to reach water temperatures up to 62°C in heat pump mode
- Working range in heat pump mode with air temperatures from -5 to 42°C.

QUALITY

- Active anode (protech) + magnesium anode
- Titanium enamelled steel tank
- Condenser wrapped around the tank

DESIGN

- LCD display
- Integrated multi-diameter adapter for air inlet and outlets
- Integrated handling straps

CE

Display easy to use, -5°C inlet air min. temperature. Renewable product

TECHNICAL DATA		200	250	250 SYS
COP air 20°C water 15-55°C (EN 255-3)		3,7	3,7	3,7
COP air 7°C water 15-55°C (EN 255-3)		3,1	3,1	3,1
COP according to EN 16147		2,61	2,8	2,8
Min/max air temperature	°C	-5/42	-5/42	-5/42
Thermal power with air at 20°C	W	2775	2775	2775
Average absorbed power (*)	W	750	750	750
Heating time with air at 20°C (*)	h,min	3,30	4,03	4,03
Heating time with air at 7°C (*)	h,min	5,12	6,04	6,04
Max. quantity of hot water at 40°C (**)	l	348	435	435
Storage tank capacity	l	200	250	250
Thermal losses in 24h	W	0,6	0,65	0,65
Max. operating pressure	bar	6	6	6
Voltage/ max. power consumption (*)	V/W	220/2500	220/2500	220/2500
Heating element power	W	1500+1000	1500+1000	1500+1000
Standard air flow rate	m^3/h	300-500	300-500	300-500
Minimum volume of installation room	m^3/h	20	20	20
Sound pressure level at 2 m	dB(A)	39	39	39
Weight when empty	kg	90	95	110
Protection rating		IPX4	IPX4	IPX4
Insulation thickness	mm	50	50	50
Diameter of hydraulic connections	"	3/4M	3/4M	3/4M
Min. temperature of storage tank room	°C	1	1	1

	200	250	250 SYS
CODE	3210031	3210017	3210018

NUOS EVO SPLIT

COMFORT

- Auto, boost, green, voyage, boost2, antilegionella functions and time setting of the withdrawals
- Coil and probe-holder for solar and/or boiler integration
- Additional double power steatite heating element

EFFICIENCY&ENERGY SAVING

- COP 3,7 with air temperature at 20°C (EN 255-3)
- COP 3,1 with air temperature at 7°C (EN 255-3)
- Ecologic gas R134A allows to reach water temperatures up to 62°C in heat pump mode
- Working range in heat pump mode with air temperatures from -5 to 42°C.

QUALITY

- Active anode (protech) + magnesium anode
- Titanium enamelled steel tank
- Condenser wrapped around the tank

DESIGN

- LCD display

CE

Super silence, -5°C inlet air min. temperature. Renewable product

TECHNICAL DATA		NUOS EVO SPLIT 300
COP air 20°C water 15-55°C (EN 255-3)		3,6
COP air 7°C water 15-55°C (EN 255-3)		2,9
COP air 7°C water 10-55°C (EN 16147)		2,71
Min/max air temperature	°C	-5/ 42
Thermal power with air at 20°C (*)	W	2450
Average absorbed power (*)	W	680
Heating time with air at 20°C (*)	H,min	5,30
Heating time with air at 7°C (*)	H,min	7,15
Max. quantity of hot water at 40°C (**)	l	549
Storage tank capacity	l	300
Thermal losses in 24h	W	0,63
Max. operating pressure	bar	6
Voltage/ max. power consumption (*)	V/W	227&2500
Heating element power	W	1500+1000
Weight when empty	kg	87
Protection rating		IPX4
Insulation thickness	mm	50
Diameter of hydraulic connections	"	3/4M
Min. temperature of storage tank room	°C	1
EXTERNAL UNIT TECHNICAL DATA		
Diameter of refrigerant connections	"	1/4 - 3/8 flare type
Weight when empty	kg	32
Standard air flow rate	m^3/h	1300
Sound power	dB(a)	60
Sound pressure level at 5 m	dB(a)	42
Max. pressure of refrigerating circuit (low-pressure side)	bar	12
Max. pressure refrigerating circuit (high-pressure side)	bar	27
Protection rating		IPX4
Max. distance between storage tank and external unit	m	8
Max. height difference between storage tank and external unit	m	3

(*) Water temperature 15-55°C with air at 20°C (EN 255-3)
(**) Water temperature 15-62°C with air at 20°C (EN 255-3)

CODE

CODE (storage tank + external unit)	3069414
Storage tank code	3069403
External unit code	3603528

AR-6PM

ENERGY EFFICIENT | ANTI FREEZING | ANTI CORROSION | SUPER SILENCE | C.O.P 4 | SINGLE COMPRESSOR

DESCRIPTION

- Compact air/water heat pump for outside installation.
- With suction gas cooled rotary compressor.
- With extensive epoxy coated hydrophilic aluminum/ rifled Cu lamellar tube evaporator and anti-corrosion coated coaxial condenser.
- Axial fan
- Refrigerant cycle with thermostatic expansion valve,
- filter, gas-liquid separator, high and low pressure switches.
- With efficient automatic defrosting by hot gas principle.
- Electric and terminal box. With control and Disturbance signaling.
- Heating regulator for mounting to the wall.
- Filled with refrigerant R417a.
- Water pump embedded for convenient installation.

		AR-6PM
Voltage	V	220
Phase		Single phase
Frequency	Hz	50
Heating capacity*	kW	5,80
Rated power input*	kW	1,36
Maximum current	A	9
Heating water capacity	L/H	124
COP*		4,26
Refrigerant		R417 A
Refrigerant charge	g	1200
Compressor		
Type		Rotary
Brand		Highly
Quantity		1
Condenser		Single wall tube in tube
Circulation pump		Grundfos
Rated water flow rate	m^3/h	1,6
Pressure Drop	Kpa	40
Circulation pump pressure head	m	2,5
Max. water temperature	°C	60
Operation temp. range	°C	-10 ~ 45
Noise	dB(A)	≤56
Connection	Inch	G3/4”
Weight	Kg	65
Index protection		IPX4
Insulation class		1
Dimension	mm	835x320x870
CODE		
		3629034

*at ambient temp. 30° C, inlet 25° C and outlet 55° C

Super silence, -10°C inlet air min. temperature. Renewable product

AR-10PM

ENERGY EFFICIENT | ANTI FREEZING | ANTI CORROSION | SUPER SILENCE | C.O.P 4 | SINGLE COMPRESSOR

DESCRIPTION

- Compact air/water heat pump for outside installation.
- With suction gas cooled rotary compressor.
- With extensive epoxy coated hydrophilic aluminum/ rifled Cu lamellar tube evaporator and anti-corrosion coated coaxial condenser.
- Axial fan
- Refrigerant cycle with thermostatic expansion valve, filter, gas-liquid separator, high and low pressure switches.
- With efficient automatic defrosting by hot gas principle.
- Electric and terminal box. With control and Disturbance signaling.
- Heating regulator for mounting to the wall.
- Filled with refrigerant R417a.
- Water pump embedded for convenient installation.

		AR-10PM
Voltage	V	220
Phase		Single phase
Frequency	Hz	50
Heating capacity*	kW	11,20
Rated power input*	kW	2,83
Maximum current	A	20
Heating water capacity	L/H	240
COP*		3,96
Refrigerant		R417 A
Refrigerant charge	g	2100
Compressor		
Type		Rotary
Brand		Highly
Quantity		1
Condenser		Single wall tube in tube
Circulation pump		Grundfos
Rated water flow rate	m^3/h	2,00
Pressure Drop	Kpa	45
Circulation pump pressure head	m	3,5
Max. water temperature	°C	60
Operation temp. range	°C	-10 ~ 45
Noise	dB(A)	≤58
Connection	Inch	G1”
Weight	Kg	84
Index protection		IPX4
Insulation class		1
Dimension	mm	1200x440x870
CODE		
		3629035

*at ambient temp. 30° C, inlet 25° C and outlet 55° C

AR-17PTP

BIG POWER | ENERGY EFFICIENT | ANTI FREEZING | ANTI CORROSION | SUPER SILENCE | C.O.P 4 | SINGLE COMPRESSOR

DESCRIPTION

- Compact air/water heat pump for outside installation.
- With suction gas cooled rotary compressor.
- With extensive epoxy coated hydrophilic aluminum/ rifled Cu lamellar tube evaporator and anti-corrosion coated coaxial condenser.
- Axial fan
- Refrigerant cycle with thermostatic expansion valve,
- filter, gas-liquid separator, high and low pressure switches.
- With efficient automatic defrosting by hot gas principle.
- Electric and terminal box. With control and Disturbance signaling.
- Heating regulator for mounting to the wall.
- Filled with refrigerant R417a.

4

		AR-17PTP
Voltage	V	380
Phase		three phase
Frequency	Hz	50
Heating capacity*	kW	21,70
Rated power input*	kW	5,31
Maximum current	A	14
Heating water capacity	L/H	465
COP*		4.09
Refrigerant		R417 A
Refrigerant charge	g	3900
Compressor		
Type		Rotary
Brand		Highly
Quantity		1
Condenser		Single wall tube in tube
Circulation pump		-
Rated water flow rate	m^3/h	4,00
Pressure Drop	Kpa	60
Circulation pump pressure head	m	-
Max. water temperature	°C	60
Operation temp. range	°C	-10 ~ 45
Noise	dB(A)	≤60
Connection	Inch	G1”
Weight	Kg	160
Index protection		IPX4
Insulation class		1
Dimension	mm	830x850x1120
CODE		
		3629036

CUTTING AREA

High efficiency, with minimum input power produced
Big power output

AR-35PTP

BIG POWER | ENERGY EFFICIENT | ANTI FREEZING | ANTI CORROSION | SUPER SILENCE | C.O.P 4 | DUAL COMPRESSOR

DESCRIPTION

- Compact air/water heat pump for outside installation.
- With suction gas cooled scroll compressor.
- With extensive aluminum/Cu lamellar tube evaporator and coaxial condenser.
- Axial fan
- Refrigerant cycle with thermostatic expansion valve, filter, gas-liquid separator, high and low pressure switches.
- With efficient automatic defrosting by hot gas principle.
- Electric and terminal box. With control and Disturbance signaling.
- Heating regulator for mounting to the wall.
- Filled with refrigerant R417a.

		AR-35PTP
Voltage	V	380
Phase		three phase
Frequency	Hz	50
Heating capacity*	kW	45,30
Rated power input*	kW	11,27
Maximum current	A	29
Heating water capacity	L/H	970
COP*		4,02
Refrigerant		R417 A
Refrigerant charge	g	3700x2
Compressor		
Type		Rotary
Brand		Highly
Quantity		2
Condenser		Single wall tube in tube
Circulation pump		-
Rated water flow rate	m^3/h	8,00
Pressure Drop	Kpa	75
Circulation pump pressure head	m	-
Max. water temperature	°C	60
Operation temp. range	°C	-10 ~ 45
Noise	dB(A)	≤62
Connection	Inch	G1-1/4”
Weight	Kg	290
Index protection		IPX4
Insulation class		1
Dimension	mm	1230x1040x1420
CODE		
		3629037

Dual compressor,
Big capacity
Hot water needs